# Homework 1

**Assigned:** Session 1, September 26, 2026

**Due:** Session 3, October 10, 2026

**Covers:** Lectures 1 and 2

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**Instructions:** Work by hand; no programming is required. State the conventions you use (qubit order, basis order, gate order) and show intermediate states rather than only final answers. Where a question asks for an explanation, two or three precise sentences are expected. Part A can be attempted after Lecture 1 and Part B after Lecture 2. Estimated total time: two to three hours. Submission arrangements are announced separately.

**Conventions:** qubits are labelled left to right, so in $|ab\rangle$ the first qubit is $a$; the two-qubit basis is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; a gate applied first is written rightmost in a matrix product. The gates are $X$, $Z$, $H$, and CNOT with the first qubit as control, as defined in the notes, and $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

## A States, measurement, and interference

Attempt after Lecture 1.

**Problem A1.** One qubit in two bases

Let $|\psi\rangle = \dfrac{\sqrt{3}}{2}|0\rangle + \dfrac{1}{2}e^{i\pi/3}|1\rangle$.

(a) Verify that $|\psi\rangle$ is normalized and give the probabilities of the outcomes 0 and 1 in a computational-basis measurement.

(b) Compute the probabilities of the outcomes $+$ and $-$ in a measurement in the basis $\{|+\rangle, |-\rangle\}$. Give exact values and decimal approximations.

(c) Replace the phase $\pi/3$ by $4\pi/3$ and repeat parts (a) and (b). Which probabilities change, and why?

(d) Replace $|\psi\rangle$ by $e^{i\gamma}|\psi\rangle$ for an arbitrary real $\gamma$ and show that none of the probabilities in (a) or (b) change.

(e) Suppose the computational-basis measurement in (a) gives the outcome 1. What is the state of the qubit afterwards, and what does a second computational-basis measurement give? What does a subsequent $\{|+\rangle, |-\rangle\}$ measurement give?

**Problem A2.** A phase made visible

Let $P(\varphi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{pmatrix}$ for a real angle $\varphi$. Consider the circuit that starts in $|0\rangle$, applies $H$, then $P(\varphi)$, then $H$, and measures in the computational basis.

(a) Write the state after each gate.

(b) Show that the probability of the outcome 0 is $\cos^2(\varphi/2)$ and the probability of the outcome 1 is $\sin^2(\varphi/2)$.

(c) For which values of $\varphi$ is the outcome certain? Identify the gate $P(\varphi)$ in those cases.

(d) Explain in two sentences why the same circuit without the final $H$ gives the outcome 0 with probability $\frac{1}{2}$ for every $\varphi$, and what this says about the role of $H$.

**Problem A3.** Superposition versus ignorance

A device outputs one qubit per use. Device 1 always outputs $|+\rangle$. Device 2 flips a fair coin and outputs $|0\rangle$ on heads and $|1\rangle$ on tails.

(a) Show that a computational-basis measurement cannot distinguish the two devices, however many times it is used.

(b) Describe a procedure that distinguishes them. If the device is in fact Device 2, compute the probability that the procedure has failed to identify it after $k$ uses.

(c) A third device outputs $|0\rangle$ or $-|0\rangle$ according to a fair coin. Can any procedure distinguish it from a device that always outputs $|0\rangle$? Justify the answer in one sentence.

# B Composite systems and entanglement

Attempt after Lecture 2.

**Problem B1.** A two-qubit state

Let $|\chi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$.

(a) Show that $|\chi\rangle$ is normalized and decide, with justification, whether it is a product state or an entangled state.

(b) Both qubits are measured in the computational basis. Give the probabilities of the four outcomes.

(c) Only the first qubit is measured in the computational basis. For each outcome, give its probability and the state of the second qubit afterwards.

(d) Only the second qubit is measured in the computational basis. Give the probabilities of its two outcomes, and explain how this is consistent with the answer to (c).

(e) Show that $|\chi\rangle = (\mathbb{I} \otimes H)\,|\Phi^+\rangle$, where $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.

**Problem B2.** Correlations in two bases, and what they do not prove

Let $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$.

(a) Both qubits are measured in the computational basis. Give the four outcome probabilities.

(b) Show that $|\Phi^-\rangle = (|+\rangle\,|-\rangle + |-\rangle\,|+\rangle)/\sqrt{2}$, and hence give the four outcome probabilities when both qubits are measured in the $\{|+\rangle, |-\rangle\}$ basis.

(c) Summarize (a) and (b) in one sentence each: are the outcomes correlated or anticorrelated in each basis?

(d) Two parties share, instead of $|\Phi^-\rangle$, a classical record prepared in advance that tells each of them what to answer if asked for a $Z$ outcome and what to answer if asked for an $X$ outcome. Show that such a record can reproduce the two tables in (a) and (b) exactly, and state precisely what the lecture's CHSH argument adds that these two tables cannot.

**Problem B3.** CHSH with other settings

For $|\Phi^+\rangle$ and measurements along directions at angles $\theta_a$ and $\theta_b$ in the $x$-$z$ plane, the lecture gives $E(\theta_a, \theta_b) = \cos(\theta_a - \theta_b)$. Let $S = E(a, b) + E(a, b') + E(a', b) - E(a', b')$.

(a) Compute $S$ for $\theta_a = 0$, $\theta_{a'} = \pi/2$, $\theta_b = \pi/3$, $\theta_{b'} = -\pi/3$. Does it violate the classical bound?

(b) Compute $S$ for $\theta_a = 0$, $\theta_{a'} = \pi/2$, $\theta_b = 0$, $\theta_{b'} = \pi/2$. Does it violate the classical bound? What does this show about the relation between entanglement and violation?

(c) For the singlet state $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ the correlator is $E(\theta_a, \theta_b) = -\cos(\theta_a - \theta_b)$. Compute $S$ with the settings of (a). Is the violation of the same size?